OVER A BASE CONTAINING THE RATIONALS THE DE RHAM COMPLEX OF A LINE BUNDLE IS QUASI-ISOMORPHIC TO THAT OF THE BASE, VIA AN EXPLICIT HOMOTOPY

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file derham.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The situation of the lemma: the bundle, the projection and the de Rham complexes. Same chapter derham.tex, the setup described above the lemma (used at proof lines 1529-1531). It fixes the morphism $S \to \Spec(\mathbf{Q})$, the line bundle situation and the complexes $\Omega^\bullet_{X/S}$ and $\pi_*\Omega^\bullet_{L/S}$ whose comparison is at stake.
- The affine local computation for a polynomial algebra over a $\mathbf{Q}$-algebra. Standard: Kaehler differentials of $A[t]$ over R and the de Rham complex of a polynomial ring (used at proof lines 1535-1543). It reduces the claim to the assertion that $\Omega^\bullet_{A/R} \to \Omega^\bullet_{A[t]/R}$ is a quasi-isomorphism, with the displayed homotopy inverse.
- Integration of polynomials with rational coefficients. Standard, used at proof lines 1544-1548. The homotopy uses the unique polynomial $\int g$ with vanishing constant term whose derivative is g, which exists because R contains $\mathbf{Q}$.

Lemma 0.1. *In the situation above, assume there is a morphism $S \to \Spec(\mathbf{Q})$. Then $\Omega^\bullet_{X/S} \to \pi_*\Omega^\bullet_{L/S}$ is a quasi-isomorphism and $H^*_{dR}(X/S) = H^*_{dR}(L/S)$.*

Proof. Let R be a $\mathbf{Q}$-algebra. Let A be an R-algebra. The affine local statement is that the map

$$\Omega^\bullet_{A/R} \longrightarrow \Omega^\bullet_{A[t]/R}$$

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is a quasi-isomorphism of complexes of R-modules. In fact it is a homotopy equivalence with homotopy inverse given by the map sending $g\omega + g'\mathrm{d}t \wedge \omega'$ to $g(0)\omega$ for $g, g' \in A[t]$ and $\omega, \omega' \in \Omega^\bullet_{A/R}$. The homotopy sends $g\omega + g'\mathrm{d}t \wedge \omega'$ to $(\int g')\omega'$ where $\int g' \in A[t]$ is the polynomial with vanishing constant term whose derivative with respect to t is g'. Of course, here we use that R contains $\mathbf{Q}$ as $\int t^n = (1/n)t^{n+1}$. $\square$